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Frozen Shoulder: Diagnosis and Treatment of Adhesive Capsulitis

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Title: Frozen Shoulder: Diagnosis and Treatment of Adhesive Capsulitis

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Abstract:

Adhesive capsulitis, commonly referred to as frozen shoulder, is a frequent cause of shoulder pain and stiffness in middle-aged adults and is associated with substantial functional limitation. Although traditionally considered a self-limited condition, growing evidence suggests that many patients experience prolonged symptoms and incomplete recovery. Diagnosis is primarily clinical, based on characteristic history and examination findings demonstrating global restriction of passive glenohumeral motion, with imaging reserved for exclusion of alternative pathology. Management is predominantly nonoperative and should be individualized according to disease stage, symptom severity, and patient comorbidities. Corticosteroid injection and physical therapy provide meaningful benefit in appropriately selected patients, while surgical intervention is reserved for refractory cases. This review summarizes the current understanding of the epidemiology, clinical presentation, diagnostic approach, and evidence-based management of adhesive capsulitis, emphasizing practical considerations to guide patient-centered care.

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Introduction

Adhesive capsulitis, commonly referred to as frozen shoulder, is a clinical diagnosis characterized by shoulder pain and a global restriction of both active and passive glenohumeral motion. The condition affects approximately 2% to 5% of the general population and is most common among adults between 40 and 65 years old[1-3]. It occurs more frequently in women and is strongly associated with systemic conditions, particularly diabetes mellitus and thyroid disease[1,4-6].

Traditionally, frozen shoulder has been described as a self-limited condition that progresses through distinct clinical stages and ultimately resolves without intervention[7,8]. However, this has been increasingly challenged by longitudinal studies demonstrating that a substantial proportion of patients experience persistent pain, stiffness, and functional limitations years after symptom onset[9,10].

These findings underscore the limitations of a uniform approach to a condition with a highly individualized clinical trajectory. Despite its prevalence, frozen shoulder remains a source of diagnostic uncertainty and therapeutic frustration for clinicians and patients. Affected individuals often experience profound functional limitations leading to a substantial impact on quality of life. An improved understanding of the condition's risk factors, natural history, and variable clinical course is therefore essential to guide timely diagnosis, set appropriate expectations, and identify patients who may benefit from more targeted or aggressive interventions.

Definitions and Classification

Adhesive capsulitis refers to a pathologic process of glenohumeral joint capsule fibrosis leading to restricted active and passive range of motion accompanied by shoulder pain[1,11]. The term "frozen shoulder" is often used interchangeably, though some authorities reserve this descriptor for the clinical presentation while using adhesive capsulitis to denote the underlying pathology. Clinicians must distinguish adhesive capsulitis from other causes of a "stiff painful shoulder," including rotator cuff tears, glenohumeral osteoarthritis, and posterior shoulder dislocation[12].

Primary (idiopathic) adhesive capsulitis develops spontaneously without an obvious precipitating event, while secondary adhesive capsulitis follows trauma, surgery, prolonged immobilization, or

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occurs in association with systemic conditions[13,14]. This distinction has prognostic implications, as secondary forms, particularly those associated with diabetes, and tend to be more resistant to treatment[4,5,9].

The classic staging system divides the disease course into three phases: freezing (painful phase with progressive stiffness), frozen (stiffness predominates with less pain), and thawing (gradual recovery of motion)[1,8,15]. Bilateral involvement may occur, either concurrently or sequentially; Patients affected in one shoulder are at risk of subsequent contralateral involvement compared to the general population[16]. In contrast, true recurrence in the same shoulder is relatively uncommon[16] (Figure 1).

[Figure 1.]

While this framework provides a useful conceptual model, clinicians should recognize its limitations. The phases do not always progress in a linear fashion, their duration varies considerably between individuals, and staging does not reliably predict response to specific interventions[8,15]. Rather than rigid adherence to staging, the framework is best used to guide intensity of intervention and set realistic expectations about the disease trajectory.

Anatomy and Pathophysiology

The glenohumeral joint is stabilized by a compliant joint capsule that permits a wide range of motion while maintaining joint congruity. Key regions include the rotator interval and coracohumeral ligament, which regulate external rotation and forward elevation, and the axillary recess. These expand during abduction to accommodate normal shoulder motion[24,25].

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Adhesive capsulitis is characterized by synovial and capsular inflammation with subsequent fibrosis, capsular thickening, and loss of elasticity[1,26,27]. These changes reduce capsular compliance and joint volume, resulting in the characteristic global restriction of both active and passive motion, with external rotation often affected earliest and most severely. As inflammation subsides and fibrotic changes predominate, pain may diminish while stiffness persists[8,26].

Systemic conditions such as diabetes mellitus are associated with increased risk and disease severity. Proposed mechanisms include altered collagen metabolism, microvascular dysfunction, and impaired tissue remodeling, though definitive causal pathways remain incompletely defined[4,5,18]. The heterogeneity of these processes likely contributes to variability in clinical course and response to treatment.

Epidemiology and Risk Factors

Systemic disease represents one of the most important risk factors for adhesive capsulitis. Diabetes mellitus is the strongest and most consistently identified risk factor, with a reported prevalence of adhesive capsulitis in 10–30% among patients with diabetes, compared to 2–5% in the general population [4,5,17,18]. Meta-analytic data suggest that diabetes confers an approximately **3–5-fold increased risk** of developing frozen shoulder, with higher rates of bilateral involvement, greater symptom severity, and a higher likelihood of prolonged or incomplete recovery [4,5,18]. Both type 1 and type 2 diabetes are affected, with risk increasing with disease duration and presence of microvascular complications[17].

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Thyroid disease, particularly hypothyroidism, has been associated with an approximately **1.5–2-fold increased risk** of frozen shoulder across large observational studies (though exact underlying mechanisms remain unknown).[6,19-21]

Other recognized risk factors include prolonged shoulder immobilization, prior shoulder surgery, and trauma[1,2,22]. Neurologic conditions, cardiovascular disease, and certain endocrine disorders have also been implicated, although their relative contributions are less well defined[2,9]. Psychological factors, including depression, have been reported with increased prevalence among affected patients, though causality remains uncertain[23].

Importantly, these risk factors influence not only disease incidence but also **clinical course and response to treatment**, underscoring their relevance for prognostication, patient counseling, and individualized management planning.

Clinical Presentation and Diagnosis

Key History

Patients with frozen shoulder typically present with insidious onset of deep-seated shoulder pain without antecedent trauma[9,28]. The pain is often described as a dull ache that becomes sharp with movement, and night pain is particularly characteristic, frequently disrupting sleep and preventing patients from lying on the affected shoulder[28,29]. As the condition progresses, patients report increasing difficulty with overhead activities, reaching behind the back (e.g., fastening a bra, tucking in a shirt), and other functional tasks requiring shoulder mobility[28,30].

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The gradual onset over weeks to months distinguishes adhesive capsulitis from acute rotator cuff tears or calcific tendinitis, which typically present more abruptly[12,28]. Patients often cannot identify a specific inciting event. The dominant arm is not preferentially affected, and bilateral involvement occurs in approximately 10-20% of cases, though rarely simultaneously[16,31].

Physical Examination

The hallmark physical examination finding in frozen shoulder is a global painful restriction of passive glenohumeral range of motion, particularly in external rotation[1,6,28]. External rotation with the arm at the side is typically the earliest and most severely affected movement, often limited to less than 25 to 30 degrees compared with normal values of 60 to 90 degrees[28,32]. Abduction is commonly restricted to less than 90 to 120 degrees, while internal rotation is reduced such that patients are often unable to reach the mid-back[28,32]. Forward flexion is usually moderately limited (Figure 2).

[Figure 2]

This pattern of restricted passive motion distinguishes frozen shoulder from rotator cuff pathology, in which passive range of motion is generally preserved despite limitations in active movement[12,28]. Although pain may contribute to apparent weakness on strength testing, true muscle weakness is typically absent, and resistance testing is often limited by pain or stiffness rather than loss of strength[28].

Imaging

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Imaging is not required to establish the diagnosis of frozen shoulder, which remains primarily clinical, but it plays an important role in excluding alternative pathology and supporting the diagnosis in select cases[6,33]. Plain radiographs of the shoulder are typically normal and should be obtained to rule out glenohumeral osteoarthritis, fracture, calcific tendinitis, or other structural abnormalities that may account for pain and stiffness[1,6,33].

Advanced imaging is not routinely indicated but may be useful in atypical presentations, cases with diagnostic uncertainty, or in patients who fail to respond to initial conservative management[33,34]. Magnetic resonance imaging (MRI) can demonstrate findings supportive of adhesive capsulitis, including thickening of the joint capsule and coracohumeral ligament, obliteration of the axillary recess, and inflammatory changes within the rotator interval[1,33,36]. However, it is important to emphasize that **MRI is primarily used to exclude alternative diagnoses** such as rotator cuff tears, labral pathology, or glenohumeral osteoarthritis, rather than to confirm adhesive capsulitis, which remains a clinical diagnosis[6,33]. These findings are most prominent in earlier stages of disease and tend to evolve as capsular fibrosis progresses[37,38].

Ultrasonography may also identify capsular thickening or reduced joint volume and offers the advantage of dynamic assessment and image-guided intervention[33,39]. However, neither MRI nor ultrasound findings are pathognomonic, and imaging abnormalities do not consistently correlate with symptom severity or functional limitation[37,38]. As such, imaging should be interpreted in conjunction with clinical findings rather than used as a standalone diagnostic tool.

Management and Treatment

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The management of adhesive capsulitis is predominantly nonoperative, with treatment selection guided by disease stage, symptom severity, patient comorbidities, and functional goals[1,6,29,30]. A stepwise approach beginning with conservative measures and progressing to more invasive interventions when necessary is generally recommended[6,29,30].

Patient Education and Expectation Setting

Patient education is a critical first step in management. Clinicians should counsel patients that although spontaneous resolution may occur, the timeline is often prolonged and recovery may be incomplete without intervention[6,9,10]. Patients should also be informed of the increased risk of developing adhesive capsulitis in the contralateral shoulder[16]. Setting realistic expectations regarding treatment duration and outcomes is essential to optimize patient engagement and satisfaction.

Pharmacologic Management

Nonsteroidal anti-inflammatory drugs (NSAIDs) are commonly prescribed for symptomatic relief, though evidence supporting their efficacy as monotherapy is limited[6,29,30]. Short-term oral corticosteroids have demonstrated benefit, with one study showing improvements in pain and function comparable to exercise therapy, though exercise was superior for improving abduction and external rotation range of motion[31]. Given the potential systemic side effects of oral corticosteroids, even at low doses, their use should be carefully considered, particularly in patients with diabetes[1,31,32].

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Intra-Articular Corticosteroid Injections

Intra-articular corticosteroid injection represents one of the most effective early interventions, particularly during the freezing phase of adhesive capsulitis[1,29,33-35]. A systematic review and meta-analysis demonstrated that intra-articular steroid injection provides significant pain relief (mean difference -1.28 cm on visual analog scale at 4-6 weeks), improved function, and increased range of motion compared to placebo, with benefits maintained up to 24-26 weeks[33]. Another network meta-analysis confirmed that intra-articular corticosteroid was associated with both statistical and clinical superiority compared to no treatment or physiotherapy in the short term[8,36].

The technique and location of injection may influence outcomes. Evidence suggests that lower doses (10 mg versus 40 mg) of corticosteroid and a rotator interval approach may provide superior pain relief and functional improvement compared to higher doses and posterior approaches[1,37]. Ultrasound guidance is not clearly superior to landmark-based injection, though it may improve accuracy[6,37]. When combined with physical therapy, corticosteroid injection provides greater benefit than either intervention alone[6,8,29,30]. Complications from intra-articular corticosteroid injection are rare and generally minor, including facial flushing (1.78%), vasovagal reactions (0.71%), and transient pain (1.07%)[33]. However, in patients with diabetes, clinicians should be aware of potential effects on glycemic control and consider alternative interventions when appropriate[1,38].

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Physical Therapy and Exercise

Physical therapy plays a central role in the management of adhesive capsulitis across all disease stages[6,26,29,39]. A structured PT program typically includes patient education, pain management strategies, joint mobilization techniques, and a graduated home exercise program[6,39]. Common elements include pendular exercises for pain modulation, assisted elevation, and stretching focused on restoring capsular mobility (Figure 3)

[Figure 3]

The UK FROST trial utilized a standardized 12-session program over 12 weeks that incorporated mobilization techniques progressively stretching into the stiff range of movement and exercises progressing from gentle pendular movements to firm stretching according to disease stage[2]. Multiple mobilization techniques have demonstrated efficacy, including Maitland technique, angular mobilization, and combined mobilizations[39]. Posterior glide mobilizations are particularly effective for restoring external rotation, while high-intensity mobilization may be less beneficial than more conservative approaches[39,40].

Proprioceptive neuromuscular facilitation (PNF) patterns have shown stronger promise over conventional capsular stretching for pain reduction and functional improvement[41]. Manual therapy combined with lidocaine injection has also demonstrated benefits for pain, range of motion, and quality of life[42]. Importantly, a recent randomized controlled trial comparing physical therapy to watchful waiting found similar functional outcomes between groups, with watchful waiting providing significantly higher value due to lower costs[43]. This suggests that

while physical therapy can be beneficial, particularly when combined with other interventions, it may not be necessary for all patients, and individualized decision-making is essential.

Hydrodilatation (Capsular Distension)

Hydrodilatation, or arthrographic distension of the glenohumeral joint, has emerged as a potentially effective intervention for adhesive capsulitis[1,44-46]. This procedure involves injecting a large volume of fluid (typically saline with corticosteroid) into the glenohumeral joint under imaging guidance to distend and potentially rupture the contracted capsule. A network meta-analysis identified capsular distension as the highest-ranking intervention for pain relief, with significant improvements in both pain and function compared to placebo[29]. A Cochrane review demonstrated that distension with saline and steroid was superior to placebo for pain (number needed to treat = 2), function (number needed to treat = 3), and range of motion at 3 weeks, with benefits maintained at 6 and 12 weeks for functional outcomes[45].

Ultrasound-guided hydrodilatation with hyaluronic acid combined with physical therapy (PT) has shown superior outcomes compared to PT by itself [1,47]. The procedure is generally safe, with the most common adverse effect being pain during and immediately after the intervention[45,46]. When combined with manipulation, hydrodilatation may provide faster pain relief and earlier improvement in range of motion compared to distension alone, though long-term outcomes appear similar. Posterior intra-articular hydrodilatation may yield the most sustained outcomes for pain reduction and range of motion improvement[48].

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Emerging and Adjunctive Therapies

Extracorporeal shockwave therapy (ESWT) has emerged as a promising treatment option, ranking highest for functional improvement in network meta-analysis[29,49]. ESWT is particularly valuable in patients with diabetes, as it avoids the potential metabolic complications associated with corticosteroids[1,38,50]. Laser therapy has also shown benefits for both pain relief and functional improvement[29,51]. Other modalities with variable evidence include acupuncture, suprascapular nerve blocks, and various physical therapy modalities[29,30,48]. Ultrasound therapy has not proven effective, while radial shockwaves and whole-body cryotherapy show some promise[39,51].

Surgical Intervention

Surgical treatment is typically reserved for patients with persistent symptoms after 6 to 12 months of conservative management, though some sources suggest waiting 9 to 12 months[1,6,29,30]. The two primary surgical options are manipulation under anesthesia (MUA) and arthroscopic capsular release, which may be performed independently or in combination[1,2,52].

In MUA, the surgeon manipulates the affected shoulder under general anesthesia to stretch and tear the contracted capsule, typically supplemented by intra-articular corticosteroid injection[2,52]. Arthroscopic capsular release involves surgical division of the contracted anterior capsule in the rotator interval under direct visualization, followed by manipulation to complete the release[2,52].

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Additional procedures such as posterior capsular release may be performed at the surgeon's discretion. The landmark UK FROST trial, which randomized 503 patients to early structured physiotherapy, MUA, or arthroscopic capsular release, found that all three interventions resulted in substantial improvements in pain and function at 12 months, with no clinically important differences between groups[2]. Mean Oxford Shoulder Scores at 12 months were 37.2 for physiotherapy, 38.3 for MUA, and 40.3 for capsular release, with all differences below the minimal clinically important threshold. Importantly, MUA was the most cost-effective option, while arthroscopic capsular release carried higher risks and costs. A subsequent randomized trial comparing MUA to arthroscopic capsular release found similar improvements in pain, range of motion, and functional scores with both procedures, with MUA demonstrating better cost-effectiveness[53]. However, arthroscopy allows direct visualization and treatment of intra-articular pathology while potentially reducing complications such as fracture, dislocation, or nerve injury that may occur with MUA[52,53].

Both surgical approaches require postoperative physical therapy supplemented by multimodal pain management to maintain and improve range of motion. Early mobilization is critical to prevent recurrent stiffness[2,52].

Stage-Specific and Patient-Specific Considerations

Treatment effectiveness is moderated by disease stage, with steroid injection combined with physical therapy providing greater benefits during the freezing phase, while joint manipulation provides more benefit in the adhesive (frozen) phase[30]. Female sex, diabetes, and the use of

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adjunctive therapies also moderate treatment effectiveness, underscoring the importance of individualized treatment planning[30]. In patients with diabetes, who experience both higher incidence and greater disease severity, special consideration should be given to interventions that avoid corticosteroids[1,4,5,38].

Conclusion

Adhesive capsulitis is a common cause of shoulder pain and stiffness that can be reliably diagnosed on the basis of characteristic history and physical examination findings, with imaging reserved for exclusion of alternative pathology. Management is predominantly nonoperative and should be individualized according to disease stage, symptom burden, and patient comorbidities, with corticosteroid injection, physical therapy, and hydrodilatation serving as effective options in appropriately selected patients. Surgical intervention is reserved for refractory cases and should be considered in the context of comparable outcomes and cost-effectiveness among available strategies.

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Figure legends:

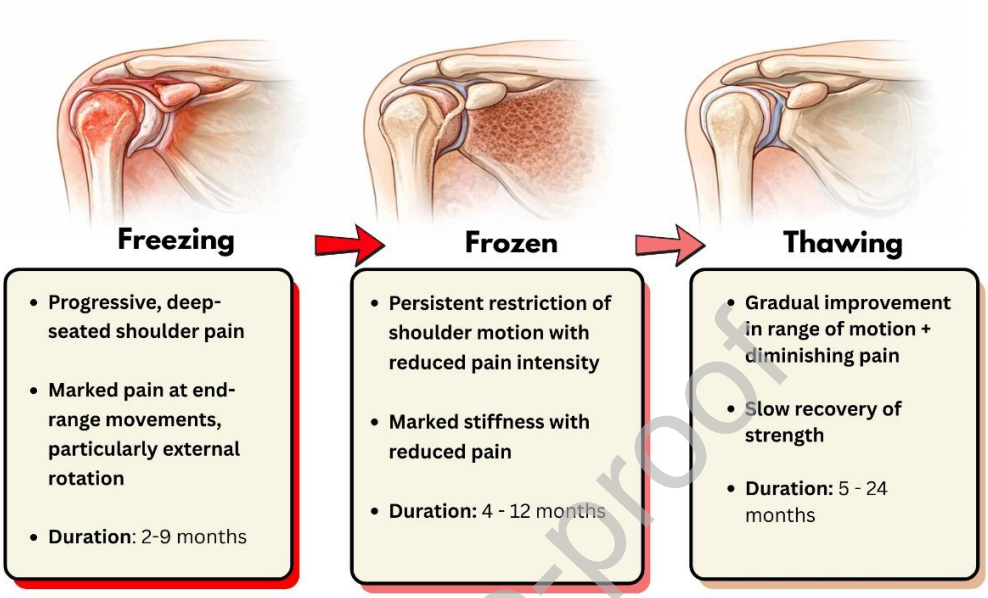
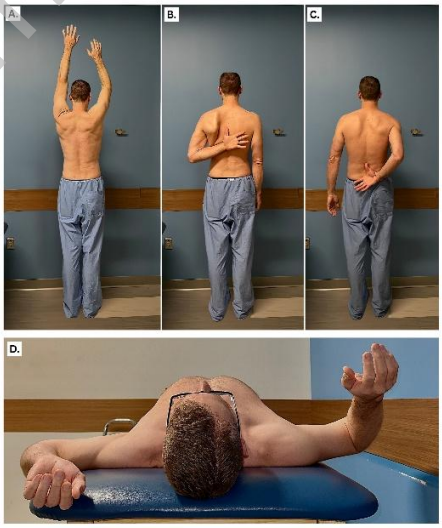


Figure 1. Clinical staging of adhesive capsulitis, classically divided into three overlapping phases: freezing, frozen, and thawing



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Figure 2. Physical examination findings in adhesive capsulitis. Representative examination maneuvers demonstrate restricted right shoulder motion, including limited overhead elevation (A), reduced internal rotation assessed by hand-behind-back reach (B, C), and marked loss of passive external rotation with the arm at the side (D).

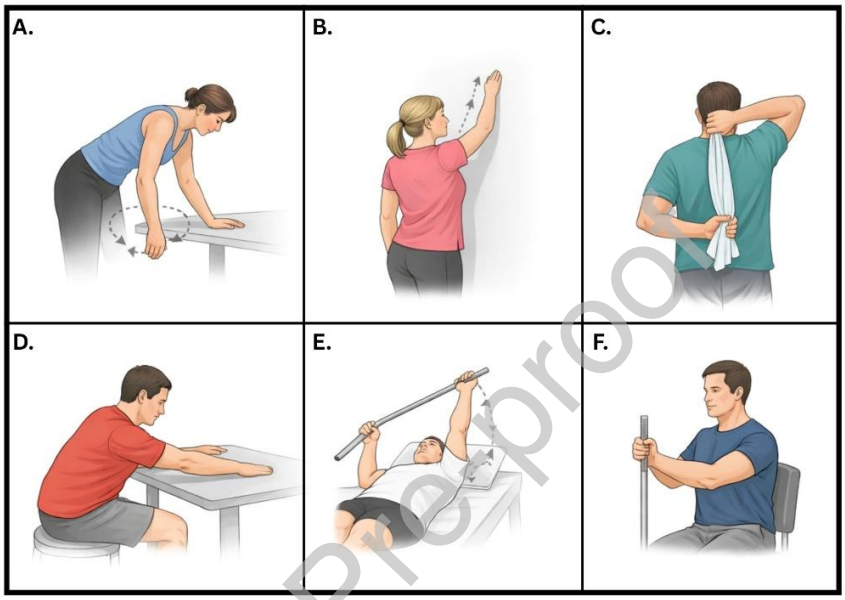


Figure 3. Representative physical therapy exercises for frozen shoulder. Examples of commonly prescribed shoulder mobility exercises are shown, including pendulum exercises for pain-modulated passive motion (A), wall-assisted elevation (B), internal rotation stretching using a towel (C), table-supported forward flexion (D), supine assisted forward elevation using a dowel (E), and seated assisted external rotation using a dowel (F). Exercises should be individualized based on symptom irritability and stage of disease, with avoidance of forceful or pain-provocative stretching.

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Clinical significance:

- Frozen shoulder is frequently not self-limited, with many patients experiencing persistent pain and functional limitation years after onset.
- Early intra-articular corticosteroid injection, particularly when combined with structured exercise, provides superior short-term pain and functional improvement compared with exercise alone.
- Routine advanced imaging adds little diagnostic value
- Patients with diabetes experience greater disease severity and poorer response to standard treatments, warranting earlier escalation or alternative strategies.

Declaration of interests

☐ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☒ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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